

P.lagopus MMRR

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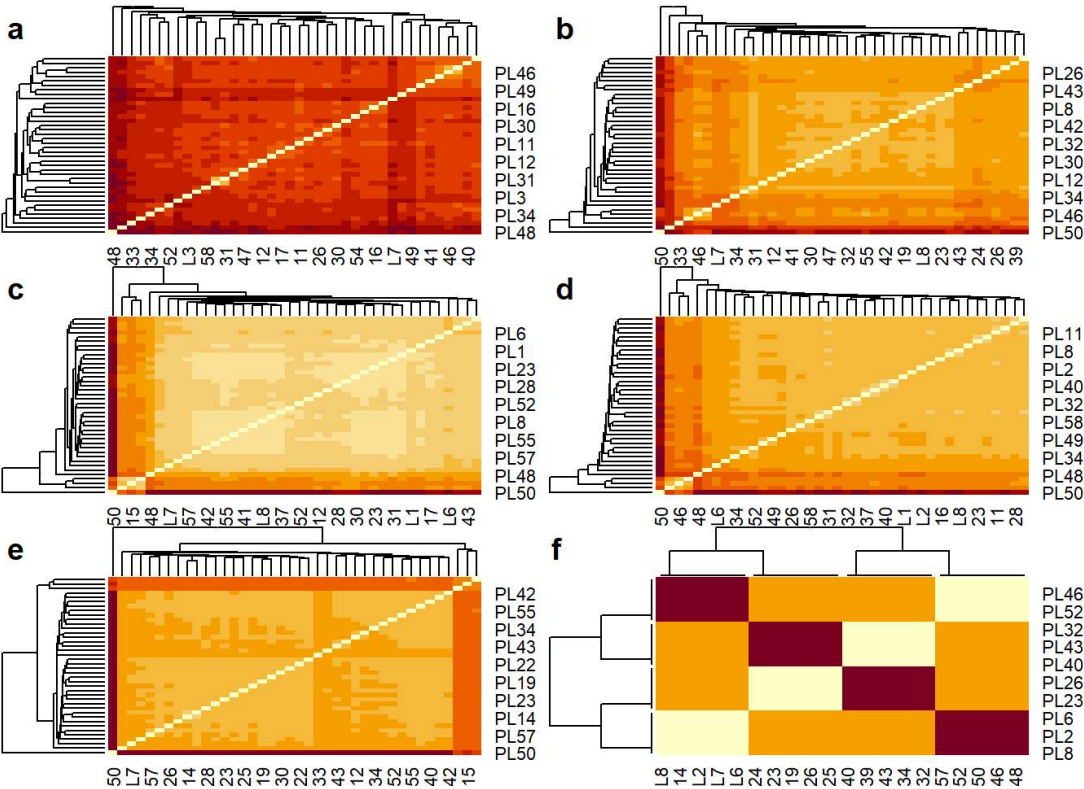


Fig. 1. Predictor distance matrices: a) CpG, b) CHG, c) CHH, d) all cytosine contexts, e) SNPs, f) Treatments.

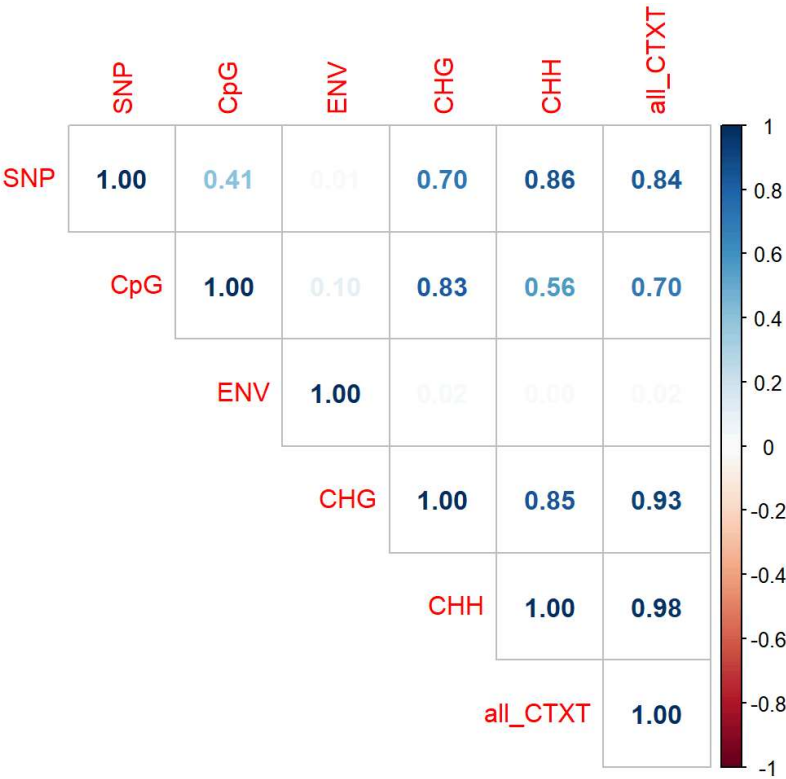


Fig. 2. Multicollinearity test for predictor selection. The predictor matrices with a correlation > 0.65 were not included in the same model.

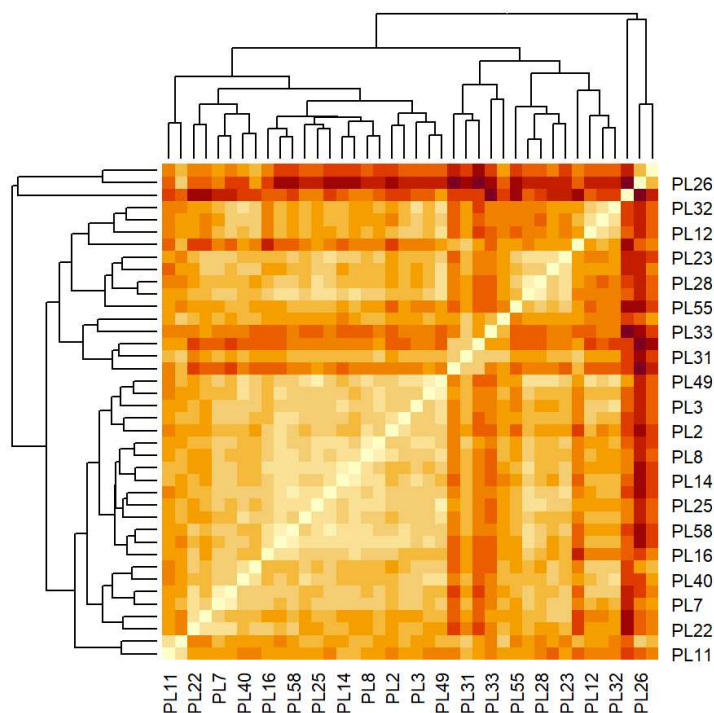


Fig. 3. Response distance matrix: Phenotypic distances based on 18 traits: morphological, chemical, phenological, physiological and reproductive.

Table 1. Results of Multiple Matrix Regression with Randomization (MMRR) showing overall model fit and individual predictor contributions for epigenetic (CpG, CHG, CHH, all contexts) and genetic models.

Model	Overall Regression				Genetic Distance		CpG Distance		CHG Distance		CHH Distance		All Contexts Distance		Environmental Distance	
	R ²	F	p-value	FDR	Coeff	p-value	Coeff	p-value	Coeff	p-value	Coeff	p-value	Coeff	p-value	Coeff	p-value
PhenD ~ CpGD + GenD + EnvD	0.051	13.898	0.163	0.443	0.177	0.178	-0.213	0.099	•	•	•	•	•	•	0.097	<0.001
PhenD ~ CHGD + EnvD	0.013	4.933	0.389	0.443	•	•	•	•	-0.083	0.524	•	•	•	•	0.077	0.001
PhenD ~ CHHD + EnvD	0.010	3.731	0.443	0.443	•	•	•	•	•	•	-0.062	0.631	•	•	0.076	0.001
PhenD ~ AllCtxtD + EnvD	0.011	4.437	0.391	0.443	•	•	•	•	•	•	•	•	-0.075	0.548	0.077	0.002
PhenD ~ GenD + EnvD	0.014	5.439	0.338	0.443	0.090	0.471	•	•	•	•	•	•	•	•	0.075	0.002

Note: Results of Multiple Matrix Regression with Randomization (MMRR). Overall model statistics include R² (proportion of variance explained), F-statistic, and associated p-value with FDR correction. Individual predictor statistics show the standardized coefficient and its p-value. ‘-’ indicates the predictor was not included in the model. Significant FDR values (p < 0.05) are shown in bold.